Overview of Ecosystem Service: Coastal Protection

This section outlines how we value coastal protection as an ecosystem service within the GEP framework. Coastal habitats reduce the energy of waves and storm surge before it reaches the shore, so property and infrastructure behind them suffer less damage than they would on a bare coast. The service is therefore valued as avoided storm and flood damage, and it is reported for two habitats: mangroves and coral reefs. Both are attributed to the country whose coast they protect, and the two components are added to give one national coastal protection value.

Methods

Estimating Quantity

Neither component is modelled from habitat extent inside this module. Both arrive as published per-country tables in which the physical protection modelling has already been done upstream, so the quantity stage here is the read and the aggregation onto the country list rather than a biophysical calculation.

Mangroves. The mangrove component is the World Bank Changing Wealth of Nations mangrove table, staged as `data_mangroves_2019.xlsx`. It carries, per country, the mangrove area in hectares, a protection value per hectare, and the annual protection value that is their product, all for 2019 and all in current US dollars. The model reads the annual value column. Because the table is keyed on the r264 country label, a country that the correspondence splits into sub-regions arrives as several rows, and those rows are summed so that each country ends up with exactly one value.

Coral reefs. The coral component is the annual expected flood protection benefit table staged as `coral_reefs_annual_expected_benefit_nfamara.xlsx`, which gives one annual benefit per country in 2011 US dollars. This table is keyed on country name rather than country code, so it is joined to the correspondence on the name and then reduced to one row per country before the currency conversion below.

Estimating Price

No price is applied in this module. Both source tables are already monetary, which means the quantity and price stages are fused upstream and the attribution factor is one:

$$\mathrm{GEP}_{\text{coastal protection}} = Q \cdot P \cdot \lambda = \text{avoided damage} \times 1 \times 1$$

The only monetary adjustment made here is a currency year alignment. The mangrove table is already in the base year's dollars and needs none. The coral table is in 2011 dollars, so each country's value is multiplied by the cumulative World Bank GDP deflator for that country over 2012 through 2019 (the annual series NY.GDP.DEFL.KD.ZG, where each year contributes a factor of one plus its rate divided by one hundred, and the eight years are multiplied together).

Valuation

The two components are joined on country and year, a country present in only one table contributes that component alone, and the national value is their sum:

$$\mathrm{GEP}_{\text{coastal protection},c} = V^{\text{mangrove}}_{c,2019} + V^{\text{coral}}_{c,2011} \cdot D_{c,2012\to 2019}$$

where $V^{\text{mangrove}}_{c,2019}$ is the country's annual mangrove protection value in 2019 dollars, $V^{\text{coral}}_{c,2011}$ is its annual coral reef protection benefit in 2011 dollars, and $D_{c,2012\to 2019}$ is its cumulative GDP deflator over that span. The result is written as one row per country for the base year 2019, with the two components kept as separate columns beside the total so that either can be inspected on its own.

On the currently staged input data this gives a global total of 36.72 billion in 2019 US dollars, of which 30.40 billion is the mangrove component and 6.32 billion is the coral reef component.

Known limitations

Three properties of the calculation are worth stating because they affect coverage rather than arithmetic, and none of them is silent once you know to look for it.

The coral table is joined on country name, not on a code. A name that the correspondence spells differently would drop the country without failing, so the join is a place to check first when a coastal country is missing.

A country the World Bank deflator series does not cover has no multiplier, so its coral value cannot be carried to the base year and the country comes out unvalued rather than valued at zero. On the currently staged data this affects nine countries holding about $75 million of 2011 coral value: the Cook Islands, Guadeloupe, Martinique, Mayotte, Niue, Pitcairn, Réunion, Taiwan and Wallis and Futuna. None of the nine carries a mangrove value either, so their rows now read as missing and the reported global total is unchanged.

Because the module consumes pre-valued national tables, it cannot separate the physical extent of a habitat from the value assigned to it. A change in either the mapped habitat area or the damage function behind these tables enters here only as a changed national total.